

JEE MAIN REVISION MASTER: STATISTICS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 124 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Statistics - Core Application	124 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Statistics - Core Application (Very High Yield — 124 PYQs)

Core Principle: Key problem pattern in Statistics.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A set of four observations has mean 1 and variance 13. Another set of six observations has mean 2 and variance 1 . Then, the variance of all..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The mean deviation about the mean for the data $\begin{matrix} x_i & 5 & 7 & 9 & 10 & 12 & 15 \\ f_i & 8 & 6 & 2 & 2 & 2 & 6 \end{matrix}$ is $\frac{a}{b}$ where a and b are coprime. Then $a+b$ is..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "If the mean of the data Class 5-10 10-15 15-20 20-25 25-30 30-35 Frequency 2 k 28 54 k + 1 5 is 21, then k is one of the roots of the equation $x^2 - 10x + 16 = 0$." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

The mean deviation about the mean for the data $x_{(i)}$ 5 7 9 10 12 15 $f_{(i)}$ 8 6 2 2 2 6 is equal to:

- (A) (40 / 13)
- (B) (42 / 13)
- (C) (44 / 13) **[Correct]**
- (D) (46 / 13)

Correct Answer: C

Solution Analysis: For a discrete frequency distribution, the mean deviation about the mean is M.D. about mean $= \frac{\sum f_{(i)} |x_{(i)} - \bar{x}|}{\sum f_{(i)}}$ First, find the mean $\bar{x}$. Step 1: Calculate $\sum f_{(i)}$ and $\sum f_{(i)} x_{(i)}$
 $\sum f_{(i)} x_{(i)}$ 5 8 40 7 6 42
 9 2 18 10 2 20 12 2 24 15 6 90 Now, $\sum f_{(i)} = 8 + 6 + 2 + 2 + 2 + 6 = 26$ and $\sum f_{(i)} x_{(i)} = 40 + 42 + 18 + 20 + 24 + 90 = 234$ So, the mean is $\bar{x} = \frac{\sum f_{(i)} x_{(i)}}{\sum f_{(i)}} = \frac{234}{26} = 9$ Step 2: Calculate $\sum f_{(i)} |x_{(i)} - \bar{x}|$ Since $\bar{x} = 9$, w